



Summer 2019 examination

# ST443

## Machine Learning and Data Mining

Suitable for all candidates

### Instructions to candidates

This paper contains five questions. Answer **ALL FIVE**.

Question 1: 30 marks

Question 2: 17 marks

Question 3: 23 marks

Question 4: 15 marks

Question 5: 15 marks

The marks in brackets reflect marks for each question.

**Time allowed -**    **Reading Time:**    *None*  
                              **Writing Time:**    *2 hours*

**You are supplied with:**    *No additional materials*

**You may also use:**    *No additional materials*

**Calculators:**    *Calculators are allowed in this examination*

1. (a) Select *True* or *False* for the following statements. Briefly justify your answers.
  - i. If the Bayes decision boundary is linear, a logistic regression classifier will attain a lower training error than a  $k$ -nearest neighbours classifier with  $k = 1$ . [2 marks]
  - ii. A decision tree generated with binary splitting has always  $m + 1$  terminal nodes, where  $m$  is the number of internal nodes. [2 marks]
  - iii. The following classifier is linear:

$$\hat{G}(x) = I(\Phi(\beta^T \mathbf{x}) > 0.2),$$

where  $\Phi(x)$  is the cumulative distribution function for the standard Gaussian distribution,  $I(\cdot)$  is the indicator function, and the class labels are 0 and 1. [2 marks]

- iv. The bagging procedure in a regression tree with  $B$  bootstrapped training sets,

$$\hat{f}_{\text{bag}}^B(x) = \frac{1}{B} \sum_{b=1}^B f^{*b}(x),$$

will overfit to the training data as  $B \rightarrow \infty$ . [2 marks]

- (b) Why does a cubic spline with, e.g., 10 degrees of freedom often give superior results to a degree-10 polynomial regression? [3 marks]
- (c) In the case of the support vector classifier, we could address the problem of a possibly nonlinear decision boundary between two classes by enlarging the feature space from  $\{X_1, X_2, \dots, X_p\}$  to  $\{X_1, X_1^2, X_2, X_2^2, \dots, X_p, X_p^2\}$ .
  - i. Develop the corresponding constrained optimisation problem based on a training dataset  $\{(x_{i1}, \dots, x_{ip}, y_i)\}_{i=1}^n$ . [5 marks]
  - ii. Discuss how the budget parameter  $C$  in the constrained optimisation problem affects the bias and variance of the support vector classifier. [3 marks]
- (d) The table below is the training data with  $n = 6$  observations of a 3-dimensional input  $\mathbf{x} = (x_1, x_2, x_3)^T$  and a qualitative output  $y$  (the colour is blue or red).

| i | $\mathbf{x}_i$ |   |   | $y_i$ |
|---|----------------|---|---|-------|
| 1 | 0              | 3 | 0 | red   |
| 2 | 2              | 0 | 0 | red   |
| 3 | 0              | 1 | 3 | red   |
| 4 | 0              | 1 | 2 | blue  |
| 5 | -1             | 0 | 1 | blue  |
| 6 | 1              | 1 | 1 | blue  |

- i. Compute the Euclidean distance between each observation in the training data and the test point  $\mathbf{x}_* = (0, 0, 0)^T$ . [2 marks]

- ii. If we use  $k$ -nearest neighbours with  $k = 3$ , what is the predicted colour corresponding to the test point  $\mathbf{x}_*$ ? [2 marks]

- (e) In class, we have shown that the standard lasso approach solves the following minimisation problem

$$\min_{\beta \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\beta\|^2 + \lambda \sum_{j=1}^p |\beta_j|,$$

where  $\lambda$  is a non-negative tuning parameter. A weighted version of the lasso approach considers solving the following minimisation problem

$$\min_{\beta \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\beta\|^2 + \lambda \sum_{j=1}^p w_j |\beta_j|,$$

where  $w_j$ 's are the known weights. Let  $\{\tilde{\beta}_j\}_{j=1}^p$  be the standard lasso or least-squares estimates. Pick a  $\gamma > 0$  and consider two possible candidates for  $w_j$  ( $j = 1, \dots, p$ ),

- $w_j = |\tilde{\beta}_j|^\gamma$ ,
- $w_j = |1/\tilde{\beta}_j|^\gamma$ .

- i. Among the above two candidate weights, which one is preferable to be used in the weighted lasso? Why? [3 marks]

- ii. What are the pros and cons of the standard lasso and the weighted lasso with the weight preferred in part i.? [4 marks]

2. Consider a two-class classification problem,  $Y \in \{0, 1\}$ . A classifier is constructed as  $\hat{G}(X) = I(\hat{p}(X) > r)$  for some user-chosen threshold  $r \in [0, 1]$ , where  $\hat{p}(X) \in (0, 1)$  is the predicted probability for  $Y = 1$  given the input  $X$  and  $I(\cdot)$  is the indicator function. The classifier is trained by minimizing the misclassification error of a training dataset  $\{(x_i, y_i)\}_{i=1}^n$  with size  $n = 1000$  for  $r = 0.5$ . The number of positive examples are  $\sum_{i=1}^n I(Y_i = 1) = 600$ . Figure 1 plots a ROC curve for the classifier based on the training dataset, that is a plot of the true positive rate (TPR) vs the false positive rate (FPR) as  $r$  varies.

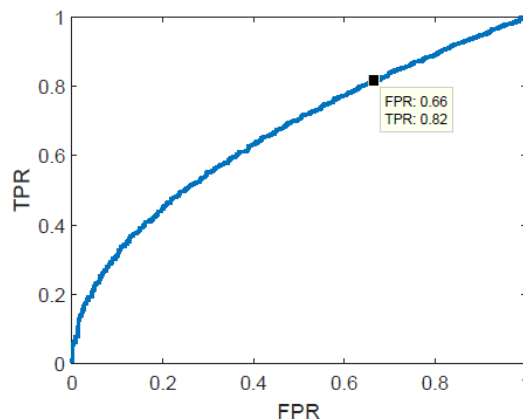


Figure 1: ROC curve for thresholds  $r \in [0, 1]$ .

- (a) A point corresponding to a specific value of  $r$  is marked with a label (FPR=0.66, TPR=0.82) in Figure 1. Construct a confusion matrix for this value of  $r$ . [6 marks]
- (b) Express the misclassification error rate as a function of FPR and TPR. [6 marks]
- (c) It just so happens that this ROC curve is well approximated by the function  $\text{TPR} = \sqrt{\text{FPR}}$ . Using this approximation, compute the FPR for the threshold value  $r = 0.5$ . [5 marks]

3. Consider the multiple linear regression model without the intercept term,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where  $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ ,  $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$ ,  $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_n)^T \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$  and  $\mathbf{X} = (\mathbf{v}_1, \dots, \mathbf{v}_p) \in \mathbb{R}^{n \times p}$  is the design matrix with  $\mathbf{v}_j = (X_{1j}, \dots, X_{nj})^T$ . The ridge regression estimator solves the following minimisation problem,

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^p} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_{j=1}^p \beta_j^2,$$

where  $\lambda$  is a non-negative tuning parameter. In class, we have shown that the solution is given by

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^T \mathbf{Y}.$$

Suppose that the columns of  $\mathbf{X}$  are mutually orthogonal.

- (a) Show that  $\hat{\boldsymbol{\beta}}_{\text{ridge}}$  can be found element by element by solving  $p$  separate ridge regression problems as follows,

$$\min_{\beta_j \in \mathbb{R}} \|\mathbf{Y} - \mathbf{v}_j \beta_j\|^2 + \lambda \beta_j^2, \quad j = 1, \dots, p.$$

[4 marks]

- (b) Show that the sum of squares explained by the full ridge regression is the sum of those explained in each of the  $p$  separate ridge regressions, that is

$$(\mathbf{X} \hat{\boldsymbol{\beta}}_{\text{ridge}})^T (\mathbf{X} \hat{\boldsymbol{\beta}}_{\text{ridge}}) = \sum_{j=1}^p (\mathbf{v}_j \hat{\beta}_{\text{ridge},j})^T (\mathbf{v}_j \hat{\beta}_{\text{ridge},j}),$$

where  $\hat{\beta}_{\text{ridge},j}$  is the  $j$ -th component of  $\hat{\boldsymbol{\beta}}_{\text{ridge}}$  for  $j = 1, \dots, p$ . [10 marks]

- (c) Let  $\mathbf{v}_1^T \mathbf{v}_1 = 2$ . Find the mean square error (MSE) for  $\hat{\beta}_{\text{ridge},1}$ . What is the value of  $\lambda$  that minimises  $\text{MSE}(\hat{\beta}_{\text{ridge},1})$ ? [9 marks]

4. This problem is about the construction of classification trees. Each split is typically made in order to minimise the cost function

$$C(T) = \sum_{m=1}^{|T|} N_m Q_m,$$

where  $T$  is the tree object,  $|T|$  is total number of terminal nodes,  $N_m$  is the number of training data points in the region corresponding to the  $m$ -th terminal node, and  $Q_m$  is the node impurity measure of the same region. In class, we have introduced two common measures including “misclassification error” and “Gini index”. Consider a two-class classification problem with two inputs and a training dataset illustrated in Figure 2. There are infinitely many possible splits we could make, but we restrict ourselves to split at integer values in the region where the training data is located.

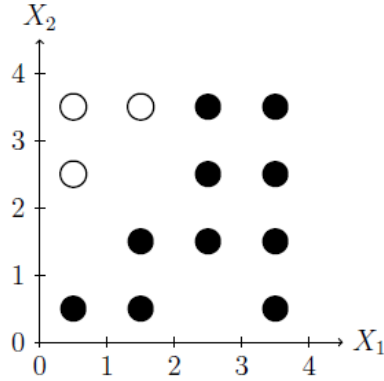


Figure 2: Training dataset with two inputs  $X_1, X_2$  and two classes for  $Y$  : “solid circle” and “hollow circle”.

- (a) Give an optimal classification tree with two split points for minimizing the training misclassification loss. [4 marks]
- (b) If we use the misclassification error as the node impurity measure, what is the resulting classification tree with two split points obtained by recursive binary splitting? Explain the suboptimality of the solution. [5 marks]
- (c) Consider the first split point. If we use the Gini index as the node impurity measure, compute the cost  $C(T)$  for the following two candidate splits
  - i.  $R_1 = \{X : X_1 \leq 1\}$  and  $R_2 = \{X : X_1 > 1\}$ ,
  - ii.  $R_1 = \{X : X_1 \leq 2\}$  and  $R_2 = \{X : X_1 > 2\}$ .

Among the above two candidate splits in i. and ii., where would we make the first split? [6 marks]

5. Let  $\mathbf{x}_1, \dots, \mathbf{x}_n$  be a dataset of  $p$ -dimensional vectors and  $\{C_1, \dots, C_K\}$  be a partition of  $\{1, \dots, n\}$ . Let  $n_k$  be the number of observations in cluster  $C_k$  for  $k = 1, \dots, K$ . For each cluster  $C_k$ , define  $\bar{\mathbf{x}}_k = \frac{1}{n_k} \sum_{i \in C_k} \mathbf{x}_i$  to be the within-cluster mean and  $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$  to

be the overall mean. Let

$$\mathbf{T} = \sum_{k=1}^K \sum_{i \in C_k} (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T \text{ to be the total deviance to the overall mean,}$$

$$\mathbf{W} = \sum_{k=1}^K \sum_{i \in C_k} (\mathbf{x}_i - \bar{\mathbf{x}}_k)(\mathbf{x}_i - \bar{\mathbf{x}}_k)^T \text{ to be the within-cluster deviance to the cluster mean,}$$

$$\mathbf{B} = \sum_{k=1}^K \sum_{i \in C_k} (\mathbf{x}_k - \bar{\mathbf{x}})(\mathbf{x}_k - \bar{\mathbf{x}})^T \text{ to be the between cluster deviance,}$$

where  $\mathbf{T}, \mathbf{W}$  and  $\mathbf{B} \in \mathbb{R}^{p \times p}$ .

(a) Verify that  $\mathbf{T} = \mathbf{W} + \mathbf{B}$ . [6 marks]

(b) Explain how the objective function of  $K$ -means is related to  $\mathbf{W}$ . [3 marks]

(c) Explain how  $\mathbf{T}$  and  $\mathbf{B}$  change during the course of the  $K$ -means algorithm. [6 marks]